# Scientific Research Experience

**A Deep Learning Enthusiast**
**Truman**
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# Introduction

985(NWAFU) EE sophomore

GPA 3.91/4.00 (92.15/100)

Really love Math&Algorithm&Physics

Deep learning Appliaction both theory

Research is my life long persuit!

**It is a pity that there are no professor doing really AI in our school, so I start my first research experiences with bioinformation,**
**but now I want to do sth**
**about CV/NLP/system**
**(AI!!!)**

# Research exprience

- 2022.6- NWAFU-LiuLab(https://github.com/NWAFU-LiuLab):

  [supervised by prof. Z liu]

  1.Sequence analysis(modification recognition by ML):

  (1) RSC (protein+machine learning )

  (2) f5C-finder (RNA+statistical learning&LLM)

  2. genomics(single cell ATAC-seq enhancement)

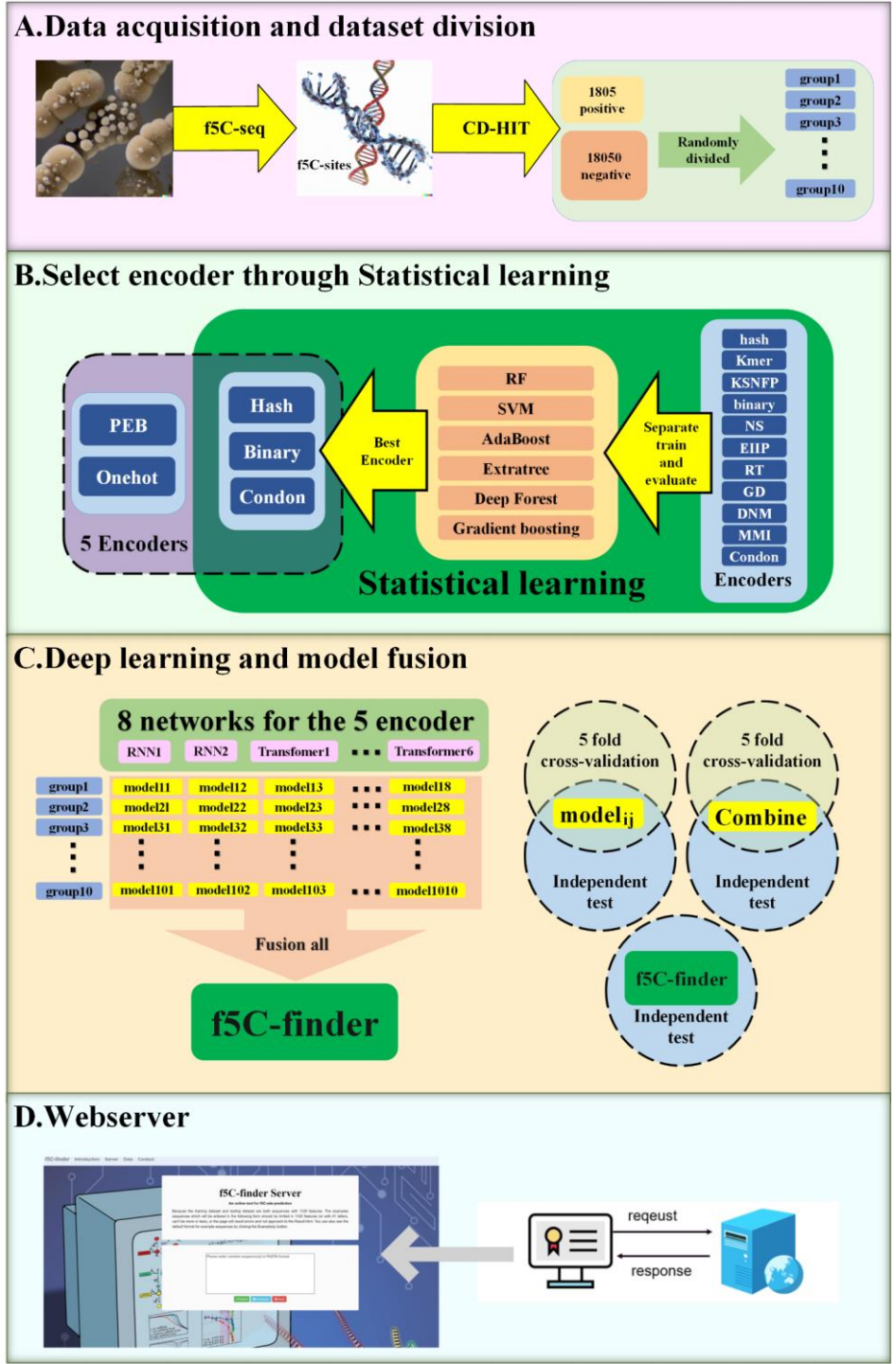
  (1) customized Swin transformer

  for Peak calling and denoise

  (2) 6mA data

# f5C-finder

1. Developed 8 LLM (2RNN+6Transformer) for the recognition of f5C modification on RNA
2. Implement the fusion model on webserver



# Skills

- **Python**
- **Pytorch&Keras (Tensorflow)**
- **Debug on linux (Shell vscode -ssh)**
- **Multi-GPU Parallel Model Training**
- **Paper writing(latex)**
- **Implement model on webserver**
- Have Debug the popular language/CV model
- Have Read the popular paper (CV/NLP/bioinformation/system)

My goal is applying CS PhD in top school
I am looking for RA opportunity
and I would self-funding to anywhere in summer vacation and do something remote in school
looking for your coopration!

# Accelerator is all I need

Anaconda Prompt (anaconda)

```
(base) C:\Users\Truman>nvidia-smi
Tue May  2 12:18:48 2023
+-----------------------------------------------------------------------------+
| NVIDIA-SMI 527.41       Driver Version: 527.41       CUDA Ver
|-------------------------------+----------------------+-------
| GPU  Name            TCC/WDDM | Bus-Id        Disp.A | Volati
| Fan  Temp  Perf  Pwr:Usage/Cap|         Memory-Usage | GPU-Uti
|                               |                      |
|===============================+======================+=========
|   0  NVIDIA GeForce ...  WDDM | 00000000:01:00.0  On |
| N/A   43C    P8    10W /  N/A |    343MiB /  6144MiB |      0%
|                               |                      |
+-------------------------------+----------------------+----------

+-----------------------------------------------------------------------------+
| Processes:                                                                  |
|  GPU   GI   CI        PID   Type   Process name                  GPU Memory |
|        ID   ID                                                   Usage      |
|=============================================================================|
|    0   N/A  N/A     24116    C+G   ...dows\System32\svchost.exe    N/A      |
|    0   N/A  N/A     85808    C+G   ...oot\Office16\POWERPNT.EXE    N/A      |
|    0   N/A  N/A     94388    C+G   E:\zotero\zotero.exe            N/A      |
+-----------------------------------------------------------------------------+
WARNING: infoROM is corrupted at gpu 0000:01:00.0

(base) C:\Users\Truman>
```

**Multi-GPU Parallel Model Training I love it!!!**

# Thanks!

```
$ nvvidia-smi
/bin/sh: 1: nvvidia-smi: not found
$ nvidia-smi
Tue May  2 12:01:54 2023
+-----------------------------------------------------------------------------+
| NVIDIA-SMI 515.86.01    Driver Version: 515.86.01    CUDA Version: 11.7     |
|-------------------------------+----------------------+----------------------+
| GPU  Name        Persistence-M| Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp  Perf  Pwr:Usage/Cap|         Memory-Usage | GPU-Util  Compute M. |
|                               |                      |               MIG M. |
|===============================+======================+======================|
|   0  NVIDIA A100-SXM...  On   | 00000000:07:00.0 Off |                  Off |
|      29C    P0    59W / 400W |   1501MiB / 40960MiB |      0%      Default |
|                               |                      |             Disabled |
|-------------------------------+----------------------+----------------------|
|                  XM...  On    | 00000000:0A:00.0 Off |                  Off |
|                 55W / 400W    |  31195MiB / 40960MiB |      0%      Default |
|                               |                      |             Disabled |
|-------------------------------+----------------------+----------------------|
|                 00-SXM...  On | 00000000:47:00.0 Off |                  Off |
|                 P0    52W / 400W |      0MiB / 40960MiB |      0%      Default |
|                               |                      |             Disabled |
|-------------------------------+----------------------+----------------------|
|                 DIA A100-SXM...  On | 00000000:4D:00.0 Off |                  Off |
|                 29C    P0    52W / 400W |      0MiB / 40960MiB |      0%      Default |
|                               |                      |             Disabled |
|-------------------------------+----------------------+----------------------|
|   4  NVIDIA A100-SXM...  On   | 00000000:87:00.0 Off |                  Off |
| N/A   61C    P0   337W / 400W |  32487MiB / 40960MiB |     96%      Default |
|                               |                      |             Disabled |
+-------------------------------+----------------------+----------------------+
|   5  NVIDIA A100-SXM...  On   | 00000000:8D:00.0 Off |                  Off |
| N/A   50C    P0   320W / 400W |  32571MiB / 40960MiB |    100%      Default |
|                               |                      |             Disabled |
+-------------------------------+----------------------+----------------------+
|   6  NVIDIA A100-SXM...  On   | 00000000:C7:00.0 Off |                  Off |
| N/A   53C    P0   324W / 400W |  32583MiB / 40960MiB |     98%      Default |
|                               |                      |             Disabled |
+-------------------------------+----------------------+----------------------+
|   7  NVIDIA A100-SXM...  On   | 00000000:CA:00.0 Off |                  Off |
| N/A   62C    P0   333W / 400W |  32499MiB / 40960MiB |    100%      Default |
|                               |                      |             Disabled |
+-------------------------------+----------------------+----------------------+

+-----------------------------------------------------------------------------+
```